

Loan Approval Prediction

Short Model Report

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Executive Summary

Recommendation. Use Logistic Regression as the case-study champion and begin deployment validation with an approval-probability threshold of **0.52**. It achieved the strongest ranking performance on the held-out test set (ROC-AUC 0.876) while remaining easier to explain and govern than the tree ensembles.

Scope. The model predicts historical loan approval from borrower features. Preprocessing handles missing values, one-hot encoding, and numeric scaling. SMOTE is applied only within training data to address class imbalance and avoid test leakage.

Model Performance

Model	Precision	Recall	F1	ROC AUC
Logistic Regression	0.862	0.882	0.872	0.876
Random Forest	0.822	0.871	0.846	0.796
Gradient Boosting	0.820	0.859	0.839	0.774
Decision Tree	0.818	0.847	0.832	0.733

Interpretation. Logistic Regression leads on every reported metric. Its precision of 0.862 means that 86.2% of applications predicted as approved were actually approved in the test data. Its recall of 0.882 means it recovered 88.2% of historically approved applications. ROC-AUC of 0.876 indicates the strongest overall ranking ability across thresholds.

Model Trade Offs

Model	Business advantage	Main limitation
Logistic Regression	Best test performance; transparent and easier to govern.	May miss complex nonlinear interactions.
Random Forest	Captures nonlinear relationships and interactions.	Less interpretable; weaker ROC-AUC in this test.
Gradient Boosting	Flexible and often strong on tabular data.	More tuning and explanation effort; lower test performance here.
Decision Tree	Rules are easy to visualize and explain.	Can be unstable and overfit small datasets.

Threshold Recommendation

Suggested threshold: 0.52. The threshold was selected from out-of-fold training predictions using an illustrative policy in which a false approval costs five times as much as a false rejection, subject to a minimum 70% recall constraint. The recall constraint prevents the cost function from selecting the commercially useless strategy of rejecting nearly every application.

Policy	Precision	Recall	F1	False approvals	False rejections
Default 0.50	0.862	0.882	0.872	12	10
Suggested 0.52	0.869	0.859	0.864	11	12

Business effect. Moving from 0.50 to 0.52 slightly improves precision and removes one false approval in the test sample, but creates two additional false rejections and slightly reduces F1. This is a conservative adjustment, not a universal cutoff.

Deployment decision. Use 0.52 only as a starting point for controlled validation. Replace the assumed 5:1 cost ratio with actual expected credit loss, loan margin, review cost, approval-volume targets, and risk appetite. A practical policy can route borderline scores to manual review rather than making every decision automatically.

Business Interpretation

Permutation importance identified the following leading predictive signals:

- Credit History was the strongest model signal.
- Property Area, Loan Amount, Married status, and Education also contributed to prediction.
- Importance represents predictive association, not causation or legal permission to use a variable.

Governance note. Gender and Married status may be sensitive or act as proxy variables. Their use must be reviewed for fairness, explainability, and applicable lending regulations before any production decision.

Business Recommendations

- Adopt Logistic Regression as the preferred decision-support model because it achieved the best ROC-AUC of 0.876 and offers clearer explanations than the tree-based alternatives.
- Begin controlled validation at a threshold of 0.52 to slightly reduce risky false approvals while maintaining strong recall.
- Route applications close to the threshold to manual review instead of automatically approving or rejecting them.
- Verify missing credit-history information because Credit History is the strongest predictive feature.
- Review Gender and Married status for fairness and regulatory compliance, then validate on larger and more recent data.
- Calibrate probabilities and monitor approval rate, defaults, overrides, fairness, and data drift after launch.

Final recommendation. Use the model to support loan officers, not to replace human judgement or established lending policy.

Conclusion

The project achieved the stated goal. It provides a reproducible pipeline for preprocessing, imbalance handling, model comparison, and cost-aware threshold selection. Logistic Regression is the preferred model, with 0.52 recommended for controlled validation. Production use requires probability calibration, recent-data testing, fairness and regulatory review, and verified business costs.